



Production and MRP Bill of Materials (BOM) Scenario

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the Bill of Materials topic in Materials Requirements Planning.
This is an advanced topic within Materials Requirements Planning so we recommend you take the Introduction to Materials Requirements Planning first.
In this course topic we frequently refer to a Bill of Materials with the acronym BOM and to Materials Requirements Planning as MRP.

Objectives



At the end of this topic, you will be able to:

- Run the MRP wizard for items with a bill of materials
- Explain MRP results for BOM items

At the end of this topic, you will be able to:

- Run the MRP wizard for items with a bill of materials.
- Explain the results that occur with different scenarios due to cumulative lead times, non-included warehouses and the inclusion of sales or assembly bills of materials.

Business Scenario



- OEC Computers produces some of their items in-house.
- These items have a bill of materials composed of inventory items, labor and resources
- Some items they sell are kits based on an assembly bill of materials.
- The procurement manager, would like to be able to plan ahead for sales demand and item availability using MRP for all these items.

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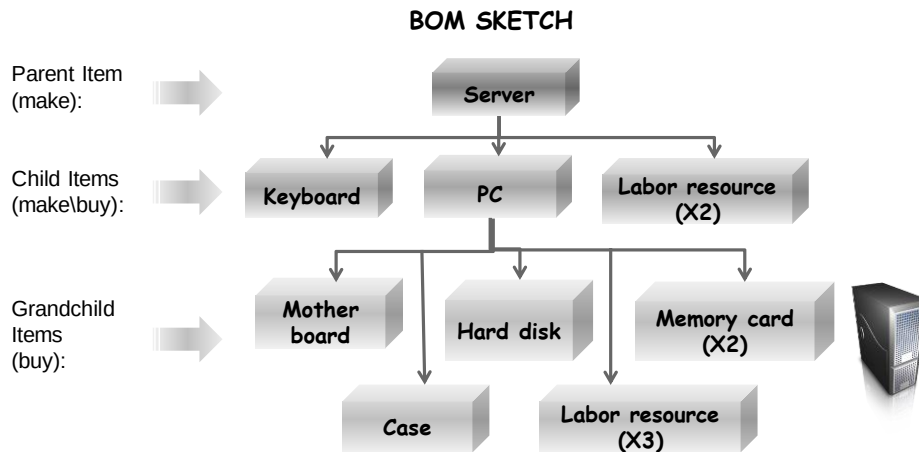
- Using MRP for BOMs: Basics
- Effects of Cumulative Lead Time
- Effects of Excluding Warehouses
- MRP for Sales or Assembly BOMs

- Concepts and Basic Scenario for using MRP with Bills of Materials
- Effects of Cumulative Lead Time
- Effects of Excluding Warehouses
- MRP for Sales or Assembly BOMs

MRP for BOMs: Basics



Example Bill of Materials



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- This graphic displays bill of materials we use in our basic scenario. This is a production BOM of a server.
- The top-level parent item is the server item. It is defined as a *Make* item in the *Item Master Data* window.
- The server is composed of a keyboard, a desktop PC and labor hours.
 - The keyboard is defined as a *Buy* item.
 - Labor is defined as a resource in the system in order to easily manage working hours quantity and costs. The resource item is not an inventory item so it is excluded from the MRP run.
 - The PC item is defined both as a *Make* item and as a BOM in the system.
 - The bill of materials is composed from the following items: a motherboard, a hard disk, 2 memory cards, a case and 3 labor hours.
 - All these item are defined as *Buy* items.

Basic Concepts for Using MRP for BOMs

When we run MRP for an item with a production bill of materials, demand for the parent item generates requirements for the child items in all BOM levels



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- Materials Requirements Planning (MRP) can be run for items produced in-house. These items are composed of items in a bill of materials that may also be produced in-house or purchased.
- When we run MRP for an item with a production bill of materials, MRP calculates gross requirements for the highest BOM level by carrying down the parent demands through the BOM structure, to its lowest levels. This means that demand for the parent item generates requirements for the children items, for all BOM levels.
- For any inventory items in the bill of materials, we can receive 3 types of recommendations:
 - Production orders for produced items,
 - Purchase documents for purchased items, and
 - Inventory transfer requests for both kinds of items (assuming we chose to receive inventory transfer request recommendations and inventory is available in other warehouses)

Running MRP for the Parent Item

Step 3 of the wizard

Item Selection

☐ All Items

☒ Selected Items

#	Item No.	Item Description	Item Procure...
1	➡ S10001	Server Point 10001	Make

parent item

MRP Report

Report			Recommendations	
☒ Preview MRP Run Results			☐ Enter Possible Problem Items	
#	Item No.	Item Description		6
1	➡ S10001	Server Point 10001	60	30
2	➡ P10009	PC Set	110	30
3	➡ C00011	Memory DDR RAM 512 MB	91	30
4	➡ C00097	Hard Disk Seagate 1 TB	50	

child items

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When we run MRP for the parent item, it will automatically produce recommendations for all the items in the bill of materials in addition to the top level item. This behavior is due to the fact that once a demand exists for the parent item, another demand is generated for the child item for every level of the BOM.

In our example, only the parent item was chosen in Step 3 of the MRP Wizard. However, all the child items appear in the MRP report along with the parent item.

Child Item Recommendations

When an MRP run includes a child item, all demand for the item is presented.

Report Recommendations **MRP Report**

☒ Preview MRP Run Results ☐ Filter Possible Problem Items

#	Item No.	Item Description	5	6	7
1	▶ S10001	Server Point 10001	60	30	20
2	▶ P10009	PC Set	110	30	
	Initial Inventory		110	30	
	Supply				
	Demand		110	30	
	Final Inventory				
3	▶ C00011	Memory DDR RAM 5	91	30	
4	▶ C00097	Hard Disk Seagate 1	50		

Pegging Information - mrp00001

Item Number: S10001
Description: Server Point 10001
Period From: 03.02.17 To: 03.02.17

Source	Type	Due Date	Quantity	Whse	Remarks
S10001	MRP Requirement	28.01.	60	01	
1311	Sales Order	01.02.	50	01	Document Source

OK

- When MRP includes a child item, all demand for the item is presented including direct demand for the item, not coming from its parent item. Recommendations will be given to all types of demand.
- Here in the graphic we see that Week 5 demand for the PC Set includes not only the MRP requirement from the parent but also sales order demand for the PC set item.

Running MRP for a child item

Report Recommendations

☒ Preview MRP Run Results ☐ Filter Possible Problem Items

#	Item No.	Item Description	7	8	9
1	▶ C00097	Hard Disk Seagate 1 TB	50		50
		Initial Inventory	10	40	20
		Supply	50		20
		Demand	20	20	40
		Final Inventory	40	20	20

MRP Report for Hard Disk Child Item

Demand

Pegging Information – Hard Disks

Item Number: ▶ C00097
Description: Hard Disk Seagate 1 TB
Period From: 25.02. TO: 03.03.

Demand Details

Source	Type	Due Date	Quantity	Unit
▶ P10009	MRP Requirement	25.02.	20	01
▶ 1312	Sales Order	28.02.	20	01

BOM requirement

Requirement not related to the BOM

- You may have items that you stock both for sale and as a child item in a bill of materials.
- If you run MRP for an item one of these items, the MRP run will also consider not only the demand generated directly for the object but also demand from its inclusion in a bill of materials.
- Here we see an MRP run for the hard disk child item. Even though the parent item was not chosen in Step 3, we still see the demand derived from the parent item.
- In the next section we will look at how lead times for child items can affect production of the parent item.

Effects of Cumulative Lead Time



Lead Time Definition



Lead Time

- Defined as time from when an item is ordered until received
- Based on either time to manufacture or time to purchase
- Used by MRP to schedule recommendations for child items
- Specified in Days in Item Master

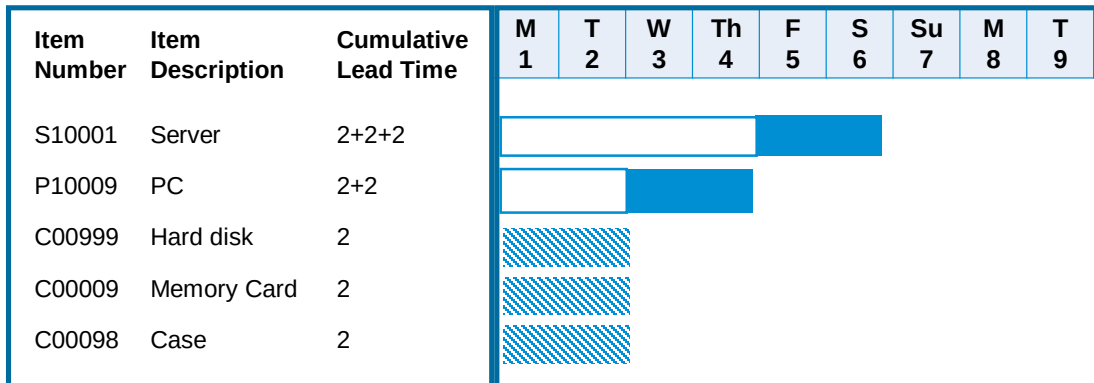
Item Master Data

Planning Data

Planning Method	MRP
Procurement Method	Buy
Lead Time	2 Days

- Lead time is the amount of time from when an item is ordered to when it is received into inventory, based on either the time to manufacture or the time to purchase. MRP uses lead time specified in the item master to determine when to schedule recommendations for child items.
- Lead time is set in the item master on the Planning Data tab.

Cumulative Lead Time Concept



Time needed to purchase



Time needed to receive components



Time needed to build

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- In our example, we need the server to be shipped to the customer on Sunday morning, so we need to receive it into stock on Saturday. Once all the components are available, the server takes 2 days to build. Therefore we need to begin to assemble the PC on Friday.
- Most components for the server are in stock, but the PC child item will need to be manufactured. The PC will take 2 days to manufacture once its components are available. Therefore, manufacturing the PC should begin on Wednesday.
- The PC has 3 child items, all of which take 2 days to procure. Therefore we need to send a purchase order to the vendor on Monday.
- When we look at the chart, we see that the cumulative lead time for the server is 6 days. 2 days for the child components of the PC, 2 days to manufacture the PC, and 2 days to assemble the server.

Options Affecting Lead Time

In Step 2 of the MRP Wizard, you have options that affect lead time:

- Consider holidays
- Ignore Cumulative Lead Time

The screenshot shows the 'MRP Wizard - Server Point' window, specifically 'Step 2 of 6: Scenario Details'. The window title is 'MRP Wizard - Server Point'. The main heading is 'Scenario Details' with the subtitle 'Define the planning horizon that you want to display in the MRP report.' The 'Description' field contains 'Server Point'. The 'Planning Horizon' section includes 'Start Date' (07.07), 'End Date' (11.10.2), 'View Data in Periods Of' (1 Weeks), and 'Planning Horizon Length' (11 Weeks). The 'Consider Holidays For' section has two checkboxes: 'Production Items' (checked) and 'Purchase Items' (checked). The 'Ignore Cumulative Lead Time' checkbox is unchecked. The 'Display Preferences' section includes 'Sort By' (Assembly Sequence), 'Display Items with No Requirements' (unchecked), 'Display Selected Items Only' (unchecked), and 'Simulation' (unchecked). At the bottom, there are buttons for 'Cancel', 'Back', 'Next', and 'Run'. The status bar at the bottom indicates 'Step 2 of 6'.

- In Step 2 (Scenario Details) of the MRP Wizard, you have a couple of options that affect lead time.
- You can decide whether to consider holidays or not. For example, you could decide to consider holidays for both production and purchase items if no production takes place during holidays and most deliveries from suppliers are also delayed due to local holidays.
- Holidays are set up from a link in the *Company Details* window.
- Another important definition is the *Ignore Cumulative Lead Time* check box. This definition is relevant for BOM items. If this checkbox is selected then the lead time will be taken from the lead time defined in the item master data. Otherwise, the lead time will be the cumulative lead times of all the recommended BOM components as shown in the previous slide.
- In the next slide, we look at what happens when you choose the option to ignore cumulative lead time.

Option: Ignore Lead Time

Item Number	Item Description	Cumulative Lead Time	M 1	T 2	W 3	Th 4	F 5	S 6	Su 7	M 8	T 9
S10001	Server	2+2+2									
P10009	PC	2+2									
C00999	Hard disk	2									
C00009	Memory Card	2									
C00098	Case	2									



Time to purchase



Time to build



Time to receive components

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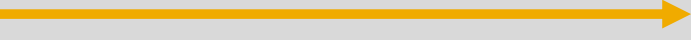
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- If you choose to ignore cumulative lead time in step 2 of the MRP Wizard, the parent item lead time is the lead time defined for it in the *Item Master Data*.
- In the image we see there is no dependencies between the different BOM levels and all items have a lead time of 2 days – as defined in the *Item Master Data*.
- Note that since the production of any parent item is dependent on the child item lead time, it is recommended not to ignore the cumulative lead time

Effect of Lead Times on Future Demand

MRP Results

Planning Horizon: **March 1 to March 30**

Report						
Item Number	Item Description	March 29	March 29	March 30	March 31	Future Data
S10001	Server					10
	Initial Inventory					
	Supply					10
	Demand					10
	Final Inventory					
P10009	PC Set				10	
C00097	Hard Disk			10		

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- Sometimes future parent demand creates MRP item component requirements within the MRP planning horizon. This can happen when the lead times cause item component demand to be within the planning horizon. In those circumstances, the parent item demand will appear in the Future Data Column.
- Recommendations that appear in the Future Data column are called theoretical recommendations. These types of recommendations only exist for items with the procurement method “make”.
- Here is an example:
 - The server is a parent item of a PC set and has requirements for April 2.
 - In the demand row of the server we see a demand for 10 server units within the future data period.
 - This demand created a recommendation within the *Future Data* column.
- In the next slide we will see how the cumulative lead time of the PC and Hard disk child items caused the recommendations to appear in the *Future Data* column.

Theoretical Recommendation: Lead Time Example

- The demand for the server is on April 2
- MRP horizon is until March 31
- Defined lead time = 1 day for each item
- Lead times are used to calculate when the first component (the hard disk) must be available to produce the PC and finally the server

Item	March 30	March 31	Future Data
Server (parent)			Demand + <i>Theoretical Recommendation</i>
PC (2nd level parent)		Recommendation	Demand (MRP Requirement)
Hard Disk (child)	Recommendation	Demand (MRP Requirement)	

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Let's look at how the demand for the server and the component lead times drive the requirements for the child items.

- The demand for the server is on April 2, outside of the MRP run's horizon. That's why the demand for the server appears in the Future Data column.
- Defined lead time = 1 day for each item
- To have the server available on April 2, MRP generates demand for the PC for April 1 (which is also in the future).
- To produce the server by April 2, MRP recommends to start by purchasing the hard disk on March 30.
- Then we need to start building the PC on March 31.
- Finally we need to manufacture the server on April 1 so it will be available April 2.

Pegging Information for Theoretical Recommendations

MRP Results

Planning Horizon: **March 1 to March 30**

Report

Item Number	Item Description	March 30	March 31	Future Data
S10001	Server			10
Initial Inventory				
Supply				10
Demand				10
Final Inventory				
P10009	PC Set		10	
C00097	Hard Disk	10		

Pegging Information - BOM

Item **S10001**

Supply Details:

Type: Production Order

Due Date: April 2

Quantity: 10

Remarks: Theoretical Recommendation

Pegging Information - BOM

Item **S10001**

Supply Details:

Type: Production Order

Due Date: April 2

Quantity: 10

Remarks: Supply from Theoretical Recommendation

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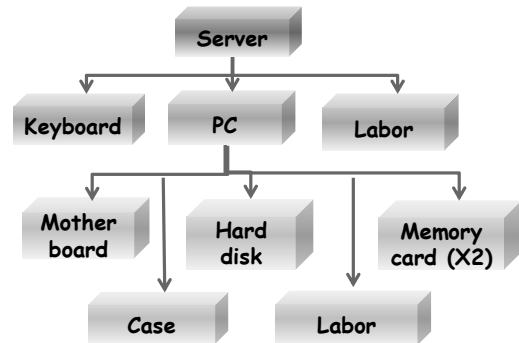
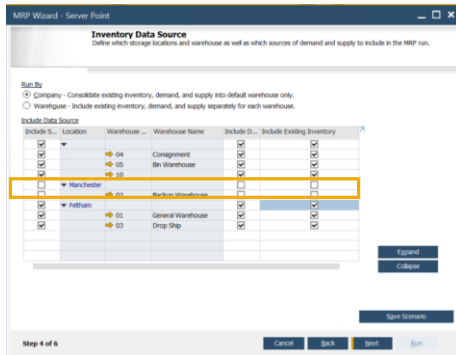
- We can see this within the MRP results by viewing the pegging information of the supply and recommendation values shown in the *Future Data* column.
- In each of the pegging information windows, we can see a remark indicating this is a theoretical recommendation.
- The purpose of this recommendation is to balance the future demand (instead of showing negative quantity).
- In the image we can see that the final inventory of the server parent item is clear.

Effects of Excluding Warehouses



BOMs and Excluding Warehouses in MRP

- You can exclude a warehouse in Step 4 of the MRP Wizard.
- Remember to ensure all child items are available in the included warehouses.



- When running the MRP wizard, you may wish to exclude a certain warehouse due to the warehouse location or other logistical considerations.
- In step 4 of the wizard you can choose which warehouse to include in the wizard run.
- When choosing this option, it is important to ensure that all child items in a BOM are available in the included warehouses.

Example: BOM and Excluding a Warehouse

Bill of Materials

Product No. P10001 X Quantity 1
 Product Description PC - P4 2.4G, DDR 512M, 400G HD
 BOM Type Production
 Production Std Cost
 Planned Average Production Size 1.00

Warehouse Price List Distr. Rule Project

Hard disk item is taken from warehouse **02** in the BOM definition

#	Type	Itc.	Description	Quantity	UoM No.	Warehouse	Issue Method	Production Std.	Total Production	Base Price
1	Item	C00001	Motherboard P4 Turbo	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price
2	Item	C00003	Quadcore P4 2.4 GHz	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price
3	Item	C00004	Tower Case with Power su	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price
4	Item	C00005	WLAN Card	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price
5	Item	C00006	Network Card 10/100	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price
6	Item	C00007	Hard Disk Seagate 400 GE	1		01	Backflush	GBP 0.00	GBP 0.00	Base Price

Warehouse **02** is not included in the scenario

Run By
☐ Company - Consolidate existing inventory, demand, and supply into default warehouse only.
☒ Warehouse - Include existing inventory, demand, and supply separately for each warehouse.

Include Data Source

Include Existing Inventory	Include Demand	Include Supply	Location	Warehouse Code
☒	☒	☒		04
☒	☒	☒	Liverpool	06
☒	☒	☒	London	05
☒	☒	☒	Manchester	02
☒	☒	☒	Feltham	01
☒	☒	☒		03

No recommendation is generated

System Message

Production orders and inventory transfer request were not recommended for 6 parent items due to components that are required from 1 warehouses whose demand has not been included in this MRP run (for example, item C00097 in warehouse 02).

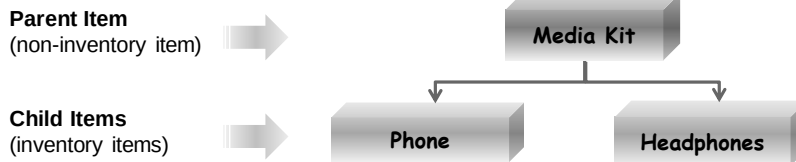
OK

- Let us examine a scenario where the warehouse defined for the child item in the BOM definition is not included in the MRP scenario.
- In the upper image we see the PC set BOM definition. Look at the hard disk child item, it is taken from warehouse **02**.
- Next look at the image of step 4 of the wizard. Note that we run the wizard by warehouse and that warehouse **02** is not included.
- With this scenario, when we run the wizard, before receiving recommendations, we receive an warning message saying no recommendations will be given to the BOM item since it cannot be produced.

MRP for Sales or Assembly BOMs



MRP for Sales / Assembly BOMs



- You can run MRP for a Sales or Assembly BOM
- Since the parent is not an *Inventory* item, it will not generate MRP recommendations
- Recommendations are given for child items only
- A compensating supply entry is added to the MRP report to balance the demand

- In addition to production BOMs, we can also run MRP for sales and assembly BOMs.
- When running the wizard for a sales or assembly BOM (that is not an inventory item), SAP Business One does not issue an actual recommendation for the parent item. It issues only the requirements for its child items (either production or purchase documents, according to the procurement method of the child items).
- Instead MRP generates a compensating entry in the supply row of the item to balance the demand and clear the final quantity.
- Recommendations for the child items are issued in the regular manner, similar to child items in a production BOM.

MRP Report for Sales and Assembly BOMs

Planning Horizon: 01.01. - 31.07.

Find Item No.:

Report: Recommendations

☒ Preview MRP Run Results ☐ Filter Possible Problem Items

#	Item No.	Item Description	Feb.	Mar.	Apr.	Jun.	Jul.	Future Data
1	M90001	Media Kit						
	Initial Inventory							
	Supply		20	20	20	20	20	20
	Demand		20	20	20	20	20	20
	Final Inventory							
2	D90001	Phone	20	20	20	40	20	
	Initial Inventory						20	
	Supply	Phone	20	20	20	40	20	20
	Demand		20	20	20	20	20	20
	Final Inventory					20		

No recommendations are given for the parent

Recommendations are given to the child item

To view the parent item, select option on MRP Step 2 :

☒ Display Items with No Requirements

- Here we see the MRP Report for our media kit.
- As we saw in the BOM hierarchy on the previous slide, the media kit is an assembly BOM with 2 items: phone and headphones.
- We do not manufacture the parent item but we want to make sure we have the media kit components on hand in time to meet the demand.
- Our sales forecast predicts that we will sell 20 media kits each month.
- However, when we run MRP we do not see recommendations for the Media Kit, only for the child items.
- Because no recommendations are given for the media kit, it will only appear in the report if we select the option in MRP Step 2 to display items with no requirements.

Assembly Sales/BOM Pegging Information

Planning Horizon 01.01. - 31.07. Calculated At 18:49

Find Item No.

Report Recommendations

☒ Preview MRP Run Results ☐ Filter Possible P

#	Item No.	Item Description	Feb.						
1	M90001	Media Kit							
	Initial Inventory								
	Supply		20	20	20	20	20	20	20
	Demand		20	20	20	20	20	20	20
	Final Inventory								

Demand came from forecasts for the parent item.

Pegging Information - mrp00001

Item Number M90001
Description Media Kit
Period From 01.02 To 28.02

Demand Details

Source	Type	Due Date	Quantity	Whse	Remarks
M2019	Forecasted Demand	01.02	20	01	Document Source

The supply came from the child items in the assembly BOM.

Pegging Information - mrp00001

Item Number M90001
Description Media Kit
Period From 01.02 To 28.02

Supply Details

Source	Type	Due Date	Quantity	Whse	Remarks
	Assembly Tree Balancing	01.02	20	01	

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- To understand the process better we can open pegging information for the supply and demand of the media kit.
- The pegging information for the demand tells us the demand was generated from a forecast.
- To balance the demand the system created a supply entry for 20 units but no recommendation.
- Look at the *Type* column of the supply pegging information. It says: *Assembly Tree Balancing*.

Summary



- Three types of recommendations are possible for items with a bill of materials
 - Production Orders for produced items (parent items)
 - Purchase Documents for purchased items (child items)
 - Inventory Requests for both types of items
- When running MRP for a parent item, demand is displayed for all child items with bills of materials
- When running MRP for a child item, demand is displayed for the item, including demand derived from the parent item
- Cumulative lead time is the total time needed to produce a bill of materials, considering the lead time of all components
- Recommendations in the Future Data column are called theoretical recommendations
- When running the wizard by warehouse level, SAP Business One does not generate recommendations when the warehouse defined for the child item is excluded from the run
- Sales and assembly BOMs can be planned with MRP but no recommendations are given for the parent item

Here are some key points to take away from this course:

- Three types of recommendations are possible for items with a bill of materials:
 - Production Orders for produced items (parent items)
 - Purchase Documents for purchased items (child items)
 - Inventory Requests for both types of items when they are coming from another warehouse.
- When running MRP for a parent item, demand is displayed for all child items with bills of materials.
- When running MRP for a child item, demand is displayed for the item, including demand derived from the parent item.
- Cumulative lead time is the total time needed to produce a bill of materials, considering the lead time of all components.
- Recommendations that appear in the Future Data column are called theoretical recommendations. These types of recommendations only exist for items with the procurement method “make”.
- When running the wizard by warehouse level, SAP Business One does not generate

recommendations when the warehouse defined for the child item is excluded from the run.

- Sales and assembly BOMs can be planned with MRP but no recommendations are given for the parent item.

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